

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

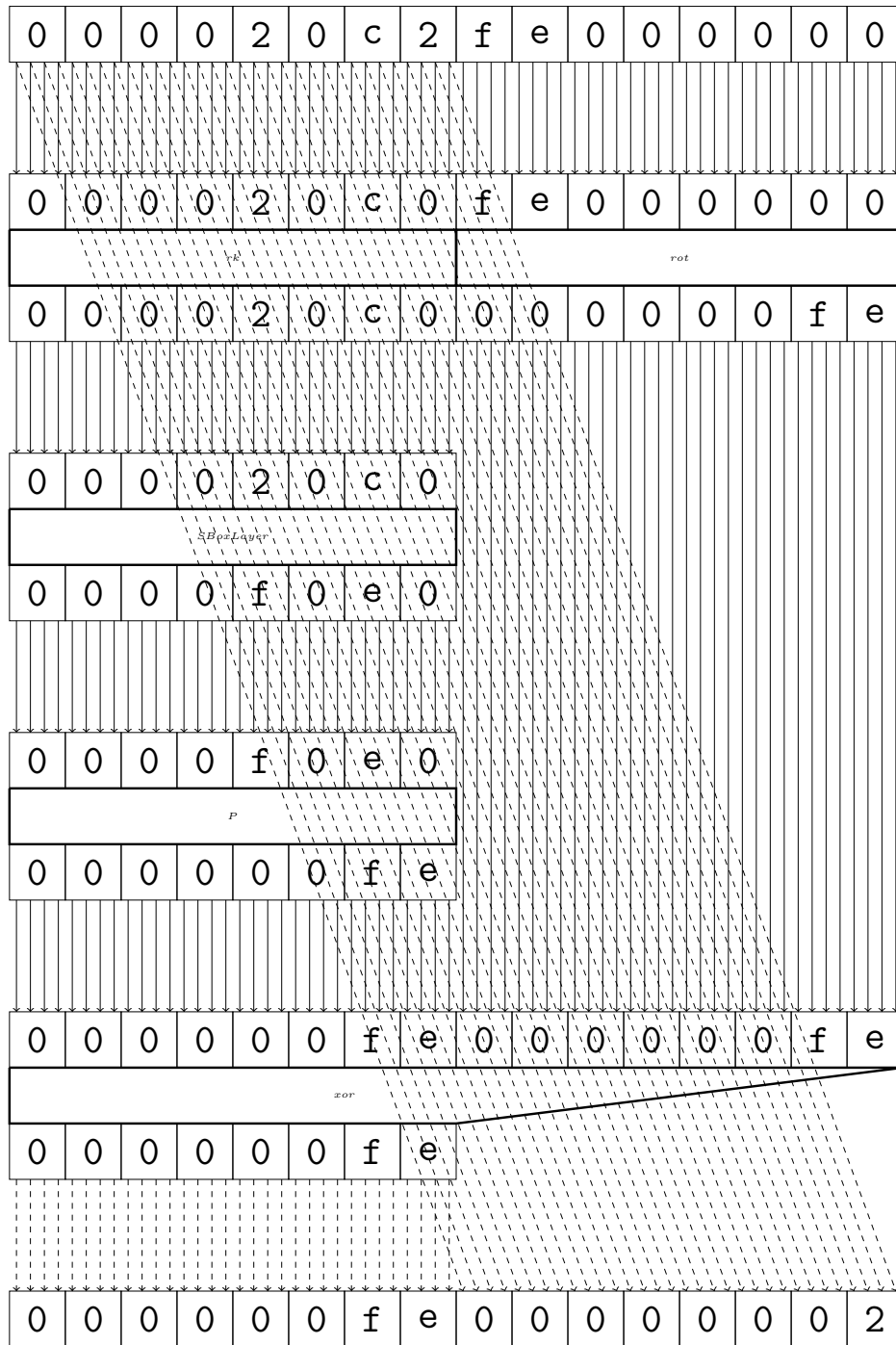
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-6}

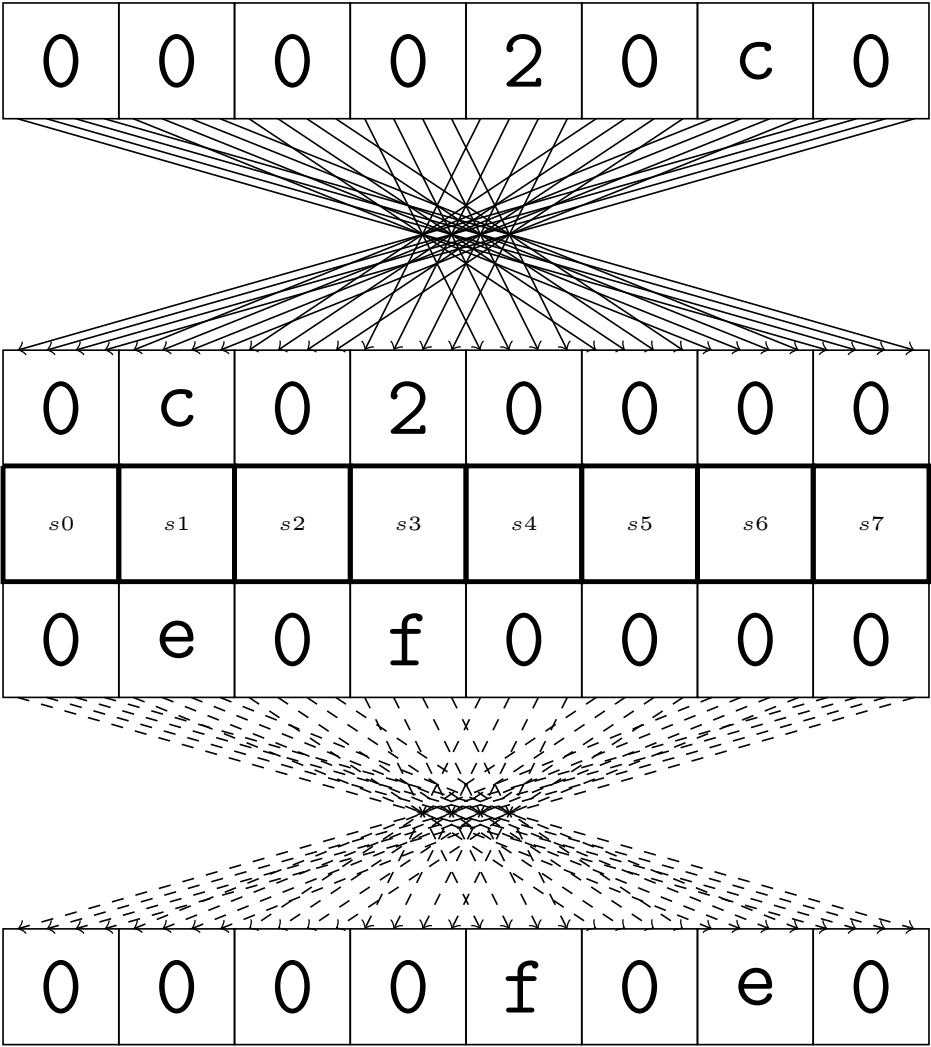
0	0	0	0	2	0	c	2	f	e	0	0	0	0	0	0
0	0	0	0	2	0	c	2	f	e	0	0	0	0	0	0
<i>LBlockRound</i>															
0	0	0	0	0	0	f	e	0	0	0	0	0	0	0	2
0	0	0	0	0	0	f	e	0	0	0	0	0	0	0	2
<i>LBlockRound</i>															
0	0	0	0	0	2	0	0	0	0	0	0	0	0	f	0
0	0	0	0	0	2	0	0	0	0	0	0	0	0	f	0
<i>LBlockRound</i>															
0	0	0	0	f	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	f	0	0	0	0	0	0	0	0	0	0	0
<i>LBlockRound</i>															
0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0
<i>LBlockRound</i>															
0	0	f	0	0	0	0	0	2	0	0	0	0	0	0	0
0	0	f	0	0	0	0	0	2	0	0	0	0	0	0	0
<i>LBlockRound</i>															
0	0	0	0	0	0	2	0	0	0	f	0	1	0	0	0
0	0	0	0	0	0	2	0	0	0	f	0	1	0	0	0
<i>FinalSwap</i>															
0	0	f	0	1	0	0	0	0	0	0	0	0	0	2	0
0	0	f	0	1	0	0	0	0	0	0	0	0	0	2	0

2 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

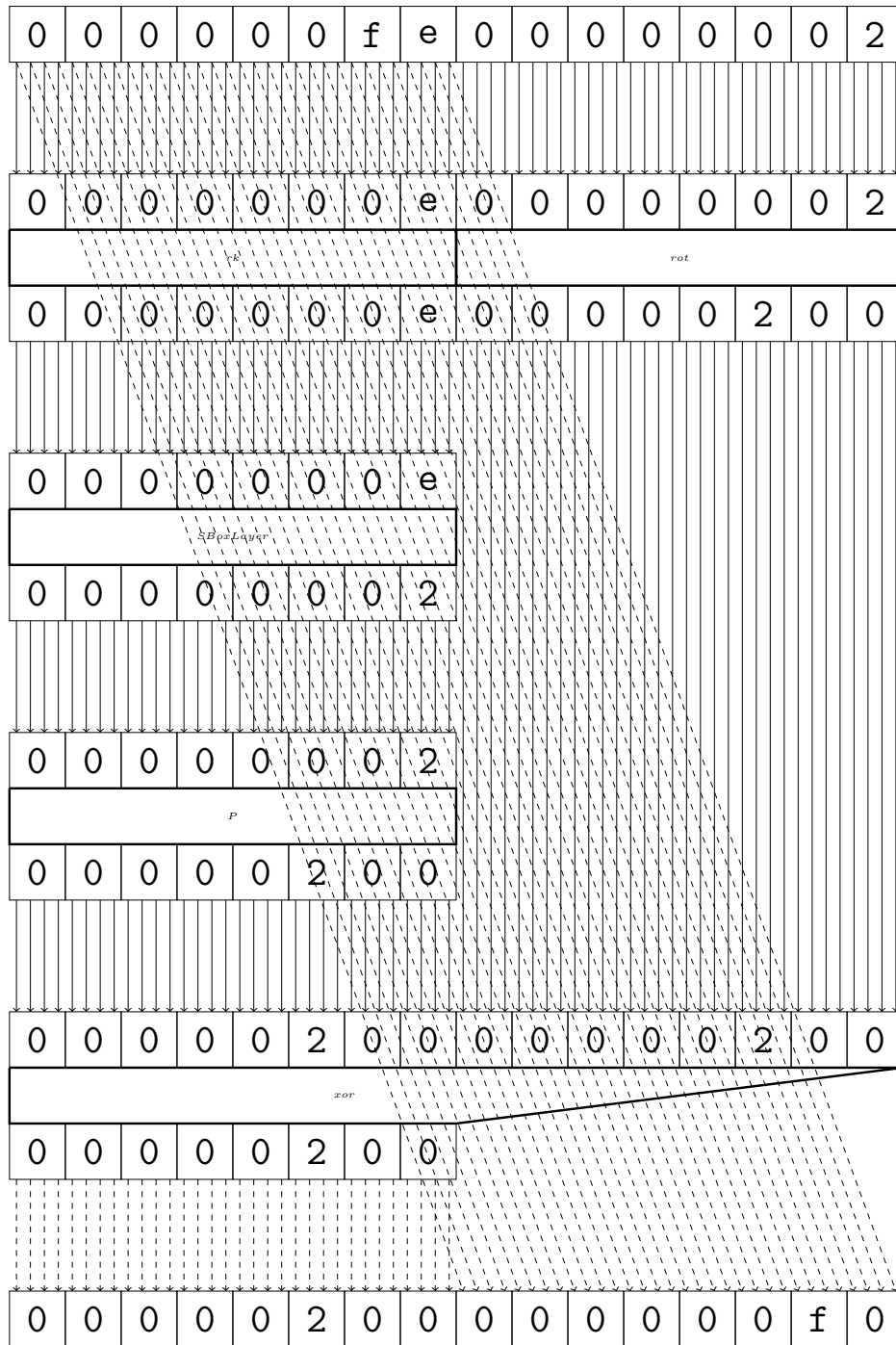
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



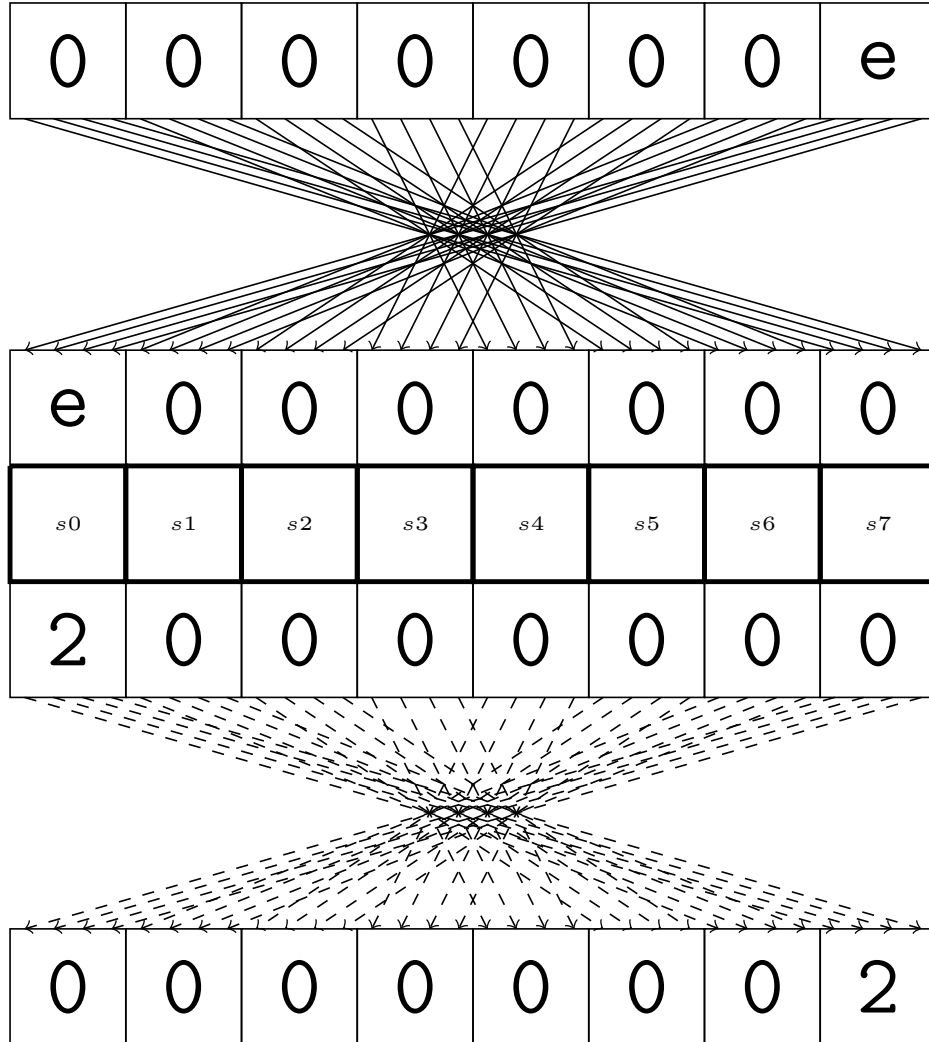
4 LBlock_round

Maximal linear correlation: $2^{-1.0}$

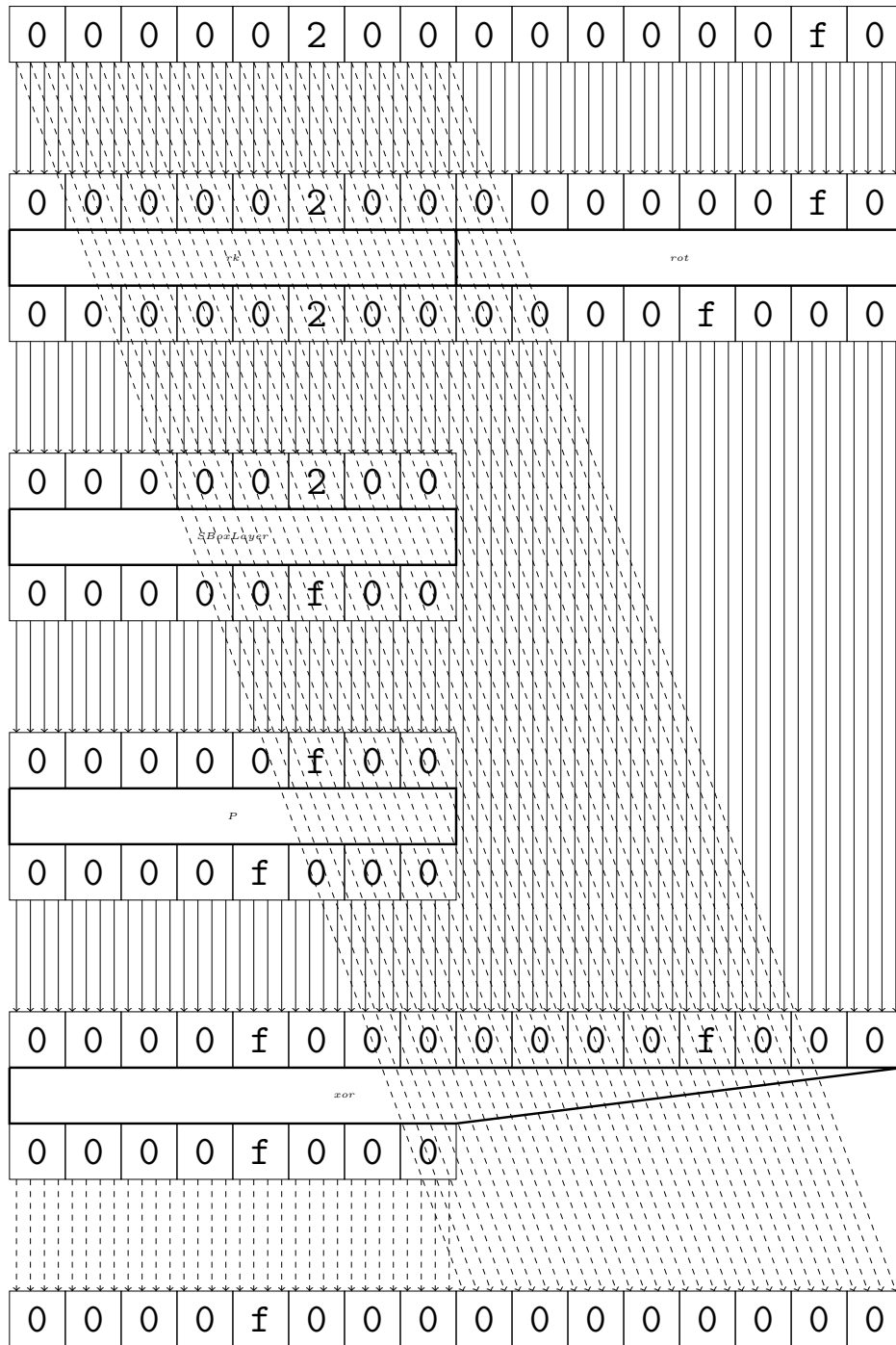


5 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

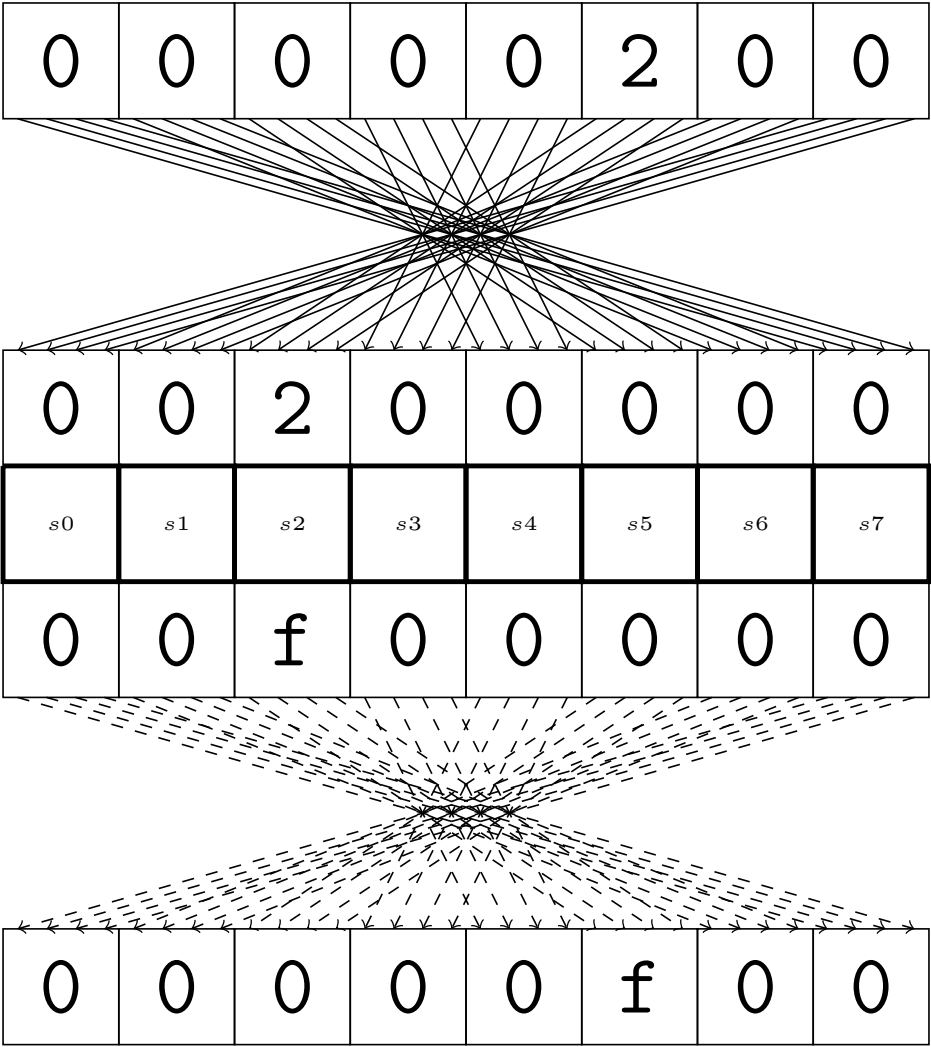


6 LBlock_round

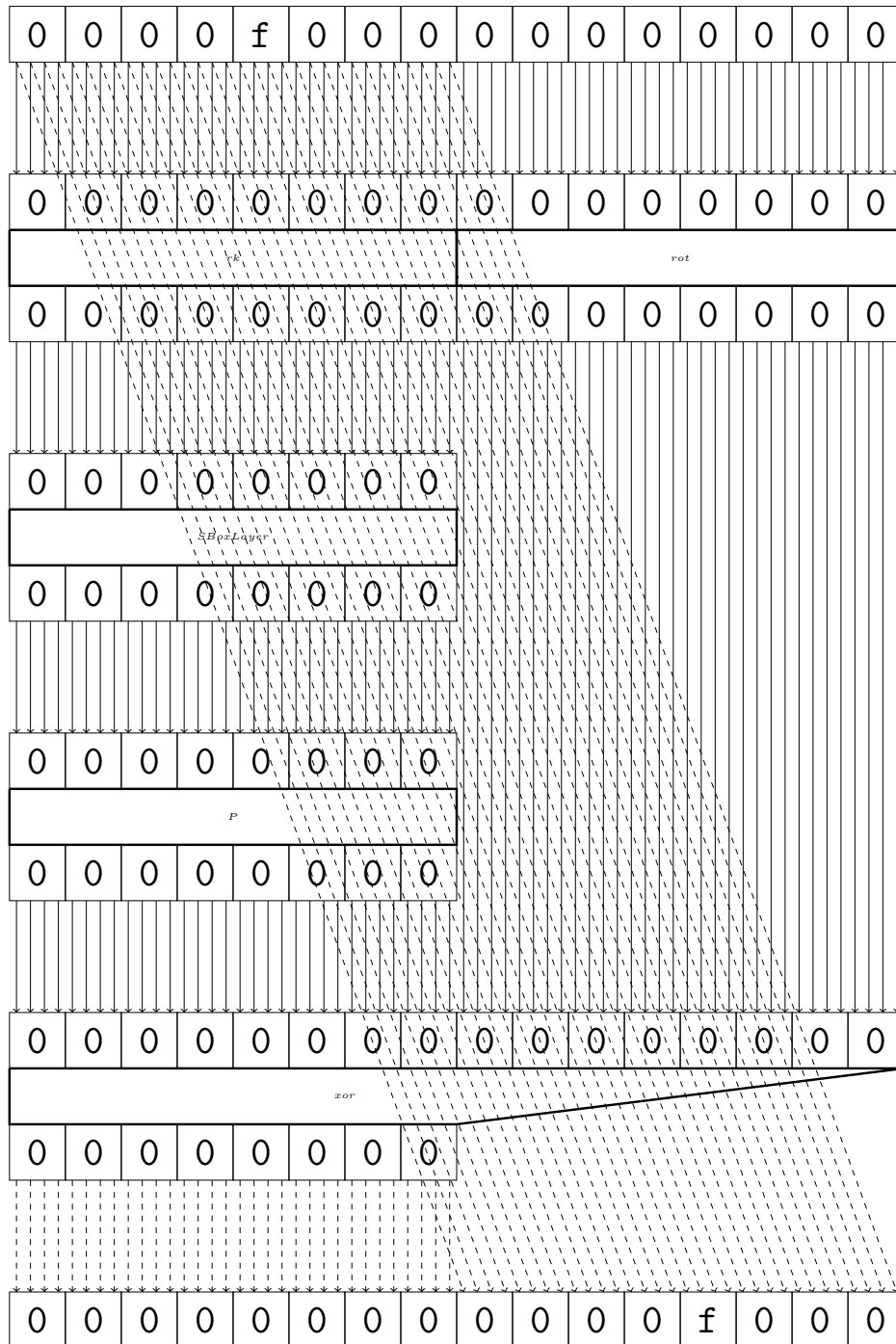
Maximal linear correlation: $2^{-1.0}$ 

7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

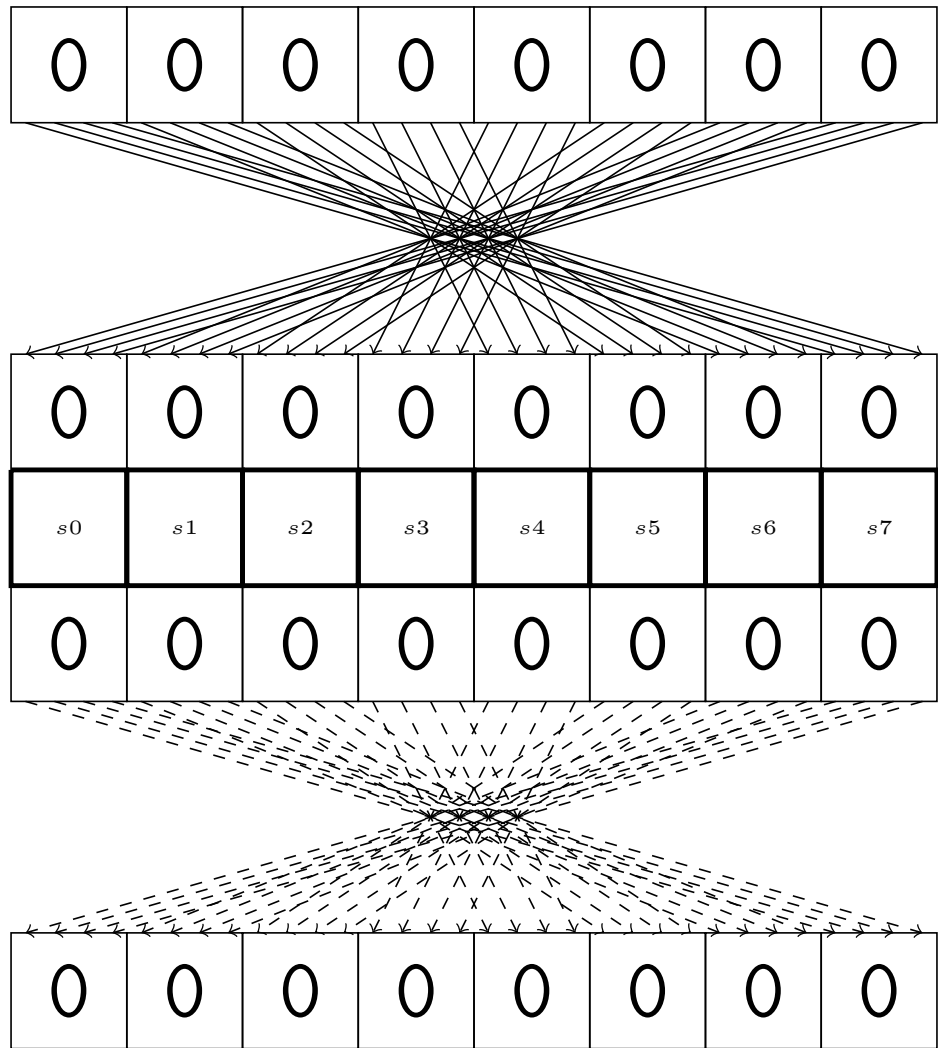


8 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

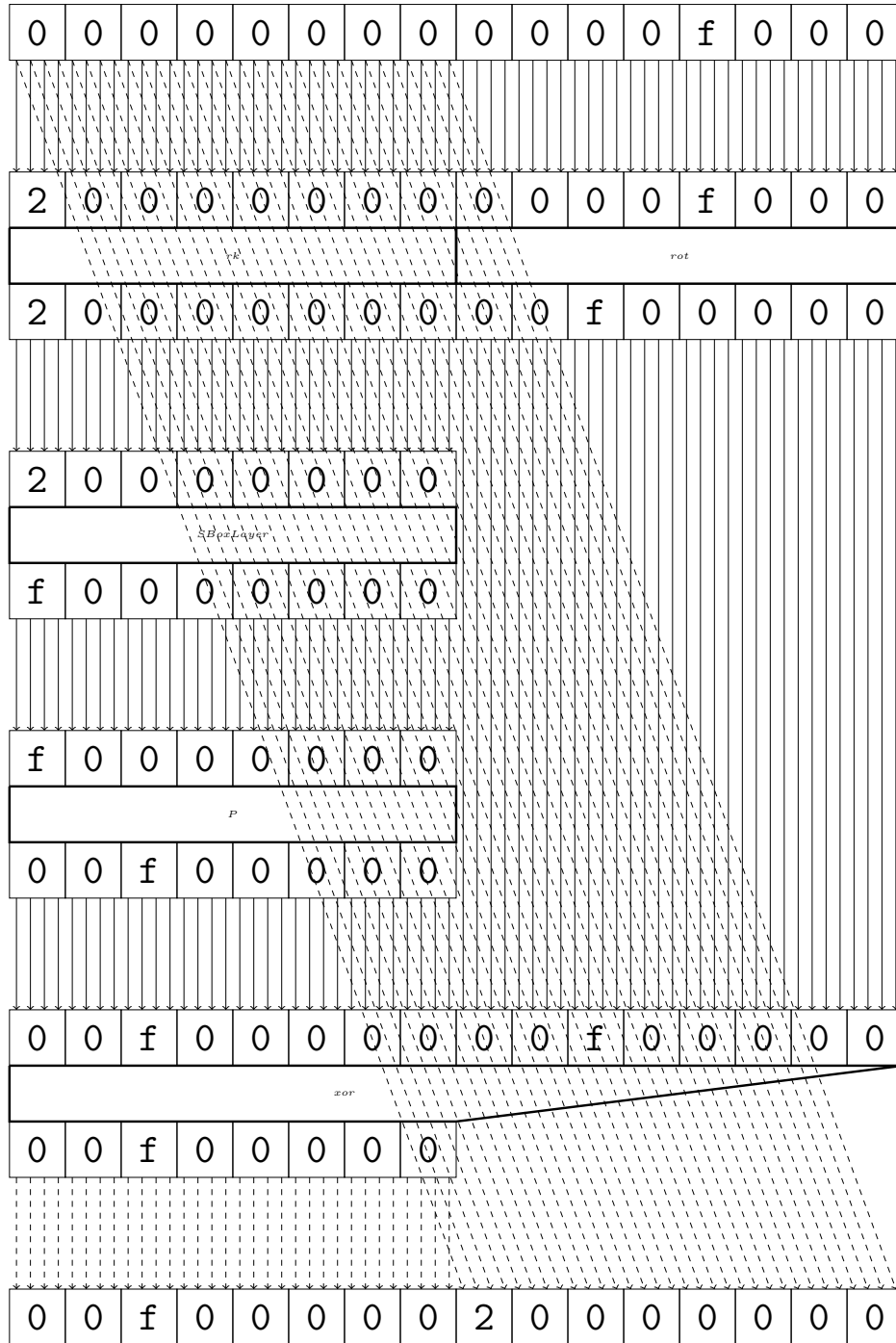
9 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



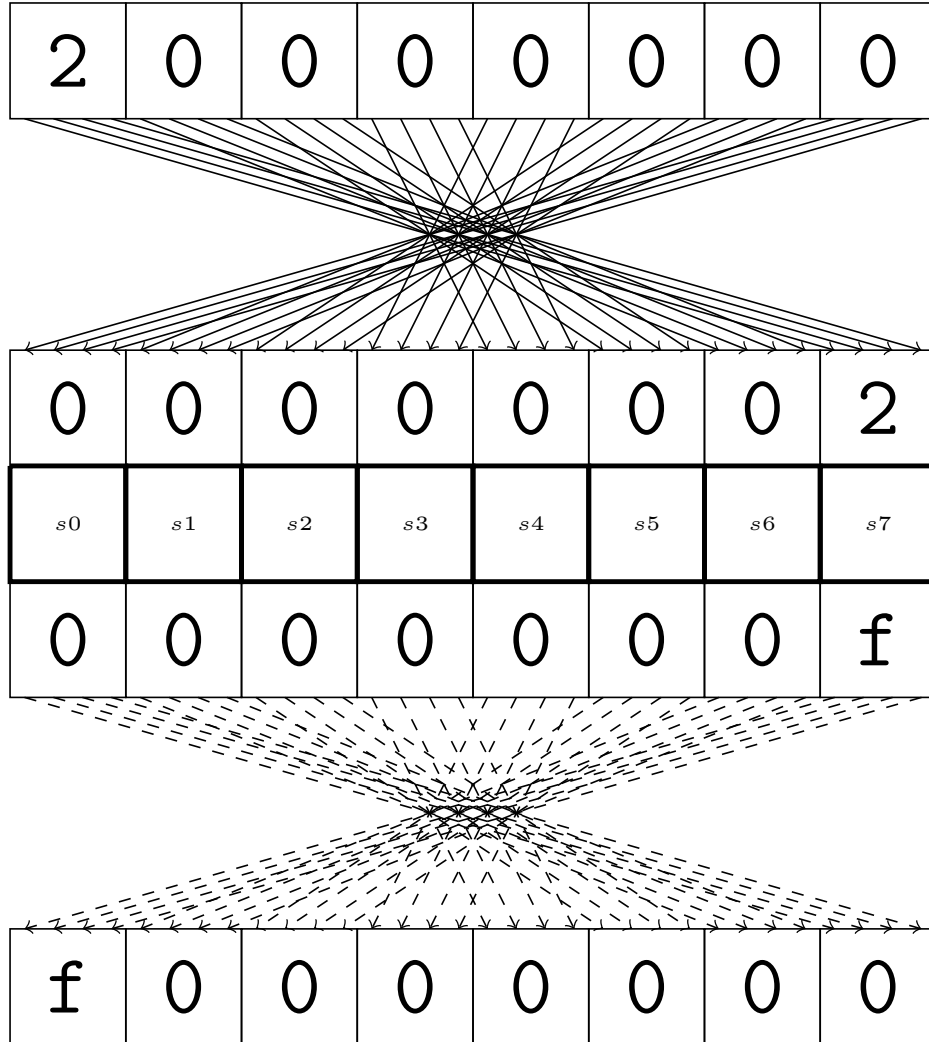
10 LBlock_round

Maximal linear correlation: $2^{-1.0}$

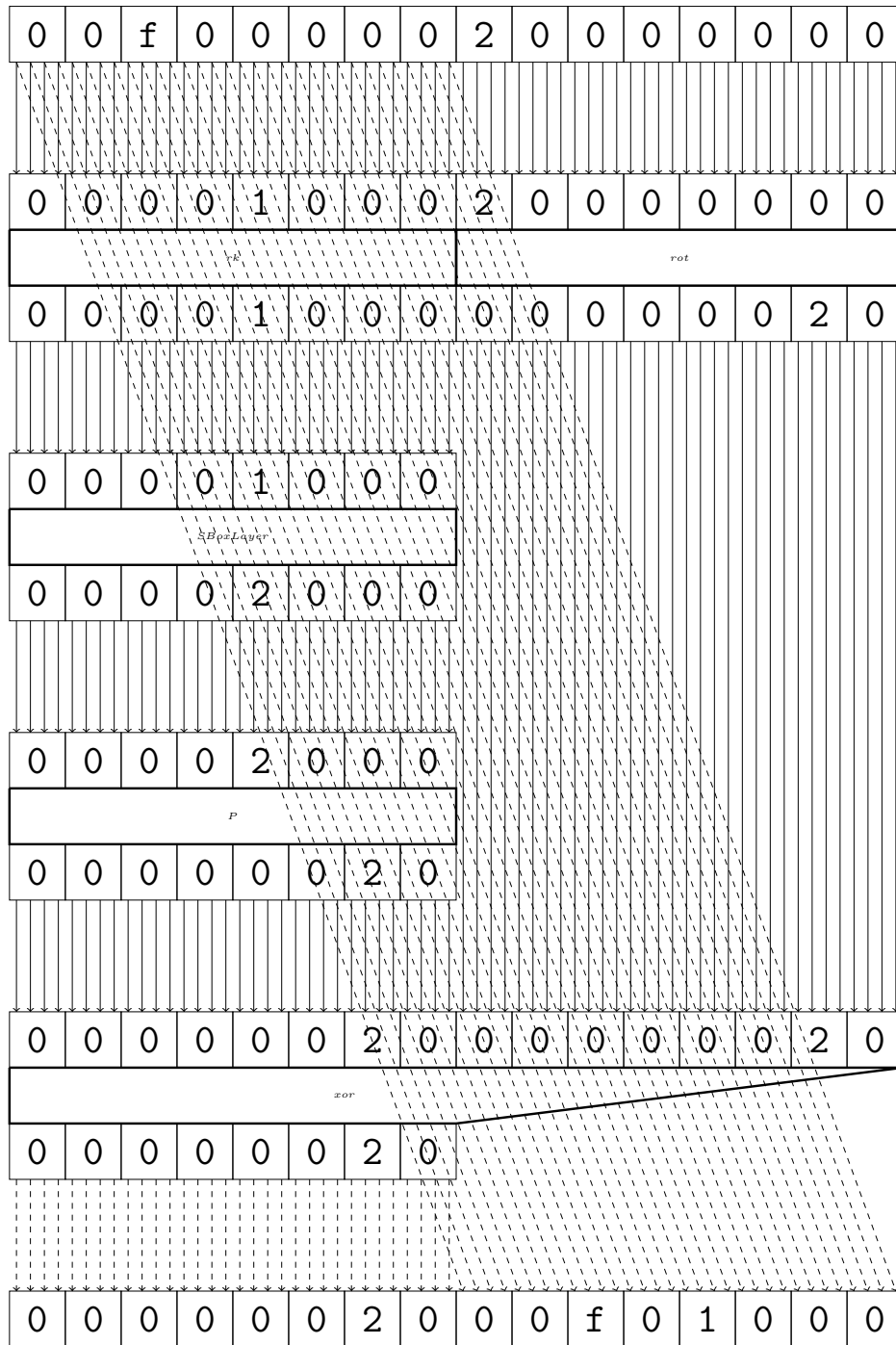


11 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



12 LBlock_round

Maximal linear correlation: $2^{-1.0}$ 

13 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

